TARMOQ XAVFSIZLIGI

A11. SSH va RDP xizmatlarini xavfsiz sozlash

1-VARIANT — Masofaviy kirishni cheklash va tekshirish

Mashg‘ulot maqsadi. Talaba Ubuntu Serverdagi SSH va Windows virtual mashinasidagi RDP xizmatlarini xavfsiz sozlaydi: server shaxsini tekshiradi, kirish huquqini alohida hisobga beradi, autentifikatsiya usulini belgilaydi va xizmatga faqat boshqaruv manzilidan ulanishga ruxsat beradi. Sozlama natijasi muvaffaqiyatli hamda rad etilgan ulanishlar bilan isbotlanadi.

Senariy. Administrator Windows 11 hostdan ikkita VMni boshqaradi. Ubuntu VMga faqat labssh foydalanuvchisi va uning SSH kaliti bilan, Windows VMga esa labrdp oddiy foydalanuvchisi orqali NLA bilan kiriladi. Uchinchi VM ruxsatsiz manba sifatida faqat ikki belgilangan portni tekshiradi.

Ko‘rsatkich	Ushbu variant
Semestr / hajm	5-semestr; A11 uchun 2 akademik soat
Vaqt	Tayyor obraz va dasturlar bilan taxminan 80 daqiqa
Xona	30 ta Windows 11 host; har talabaga mustaqil virtual lab
Host	8 yadroli CPU, 32 GB RAM, 1 TB SSD; LAN va Internet
SSH-SRV	Ubuntu Server 24.04 LTS; 2 vCPU, 2 GB RAM, 20 GB disk
RDP-SRV	Windows 11 Pro/Education/Enterprise; 2 vCPU, 6 GB RAM, 64 GB disk
TEST-VM	Ubuntu Server 24.04 LTS; 1 vCPU, 1 GB RAM, 15 GB disk
Natija	Konfiguratsiya, 8 ta asosiy sinov, loglar va 10 ballik himoya

Muhim tushunchalar

- SSH foydalanuvchi kaliti foydalanuvchini, server host key esa serverni aniqlashga xizmat qiladi. Bu ikki kalitning vazifasi boshqa-boshqa.
- RDP NLA (Network Level Authentication) foydalanuvchini to‘liq masofaviy sessiya ochilishidan oldin autentifikatsiya qiladi. NLAning o‘zi MFA emas.
- Firewall xizmatga qaysi manbadan yetib borishni boshqaradi; foydalanuvchiga kirish huquqi xizmat va OT siyosatida alohida tekshiriladi.
- TLS yoki SSH shifrlashi serverning haqiqiyligini tekshirish ehtiyojini yo‘qotmaydi. Kalit yoki sertifikat o‘zgarsa, sababini tekshirish kerak.

Chegara. Barcha sinovlar 10.110.10.0/24 host-only tarmog‘ida o‘tkaziladi. RDP xizmatini jismoniy Windows hostda yoqmang. SSH/RDPni xona LANiga yoki Internetga port forwarding orqali chiqarmang. Parol va shaxsiy kalit hisobotga qo‘shilmaydi.

1. Muhitni tayyorlash va manzillash

1.1. O‘qituvchi uchun darsdan oldingi tayyorgarlik

80 daqiqalik reja OT o‘rnatilgan va yangilangan, guest tools, shared folder hamda kerakli mijoz dasturlari ishlaydigan obrazlarga tayanadi. Windows 11 Home RDP server bo‘la olmaydi; RDP-SRV uchun Pro, Education yoki Enterprise obrazini tayyorlang. Windows host Home bo‘lsa ham RDP mijozi sifatida ishlashi mumkin. [1]

```
# SSH-SRV: vaqtincha NAT bilan, darsdan oldin.
sudo apt update
sudo apt install -y openssh-server ufw
# TEST-VM:
sudo apt update
sudo apt install -y netcat-openbsd iproute2 iputils-ping
```

Windows hostda PowerShell orqali Get-Command ssh,ssh-keygen,mstsc ni tekshiring. OpenSSH Client yo‘q bo‘lsa Settings → System → Optional features orqali o‘rnating; bu ish darsdan oldin bajariladi. Windows RDP-SRVda konsoldan foydalaniladigan administrator hisobi bo‘lsin. U domen/MDM orqali boshqarilmaydigan o‘quv VM bo‘lsin.

1.2. Host-only tarmoq

Tugun	IP / maska	Vazifa
Windows host (ADMIN)	10.110.10.1/24	SSH va RDP mijoz; yagona ruxsatli manba
SSH-SRV	10.110.10.10/24	22/TCP OpenSSH
RDP-SRV	10.110.10.20/24	3389/TCP RDP
TEST-VM	10.110.10.30/24	Ruxsatsiz manbadan port sinovi

- VirtualBox: Tools/Network Manager → Host-only Networks. Bitta adapter yarating, host IPni 10.110.10.1/24 qiling, DHCPni o‘chiring; uch VMni shu adapterga ulang.
- VMware: Virtual Network Editor → alohida Host-only VMnet; subnet 10.110.10.0/24, host virtual adapteri yoqilgan, DHCP o‘chiq. Windows virtual adapter IPsi 10.110.10.1 ekanini tekshiring.
- VMlarning NAT kabelini uzing, Bridged adapter qo‘shmang. Hostda Internet Connection Sharing yoki network bridge bo‘lmasin. Host-only tarmoq jismoniy hostni ham o‘z ichiga olishini yodda tuting.
- Windows hostdagi A11 papkasini SSH-SRV uchun A11 nomli shared folder sifatida tayyorlang. U faqat ochiq kalit va dalil almashish uchun ishlatiladi.
- Har VMdan A11-clean snapshot oling. Hostda mavjud SSH kalitlari va known_hosts faylini o‘zgartirmaslik uchun alohida A11 fayllari ishlatiladi.

Uch VM uchun jami taxminan 9 GB RAM yetadi. 30 hostda bir xil manzillar qo‘llanishi mumkin, chunki host-only tarmoqlar o‘z hosti doirasida qoladi. O‘qituvchi darsdan oldin bu subnet xona LANi yoki VPN bilan to‘qnashmasligini tekshiradi.

1.3. VMlarning IPv4 manzillarini sozlash

Ubuntu VMlarda ip -br link bilan lab interfeysini MAC bo‘yicha aniqlang. Quyidagi enp0s8 namunaviy; haqiqiy nomni ishlating. Buyruqlar tegishli VM konsolida bajariladi; shlyuz va DNS talab qilinmaydi.

```
# SSH-SRV:
sudo -i
export TZ=UTC
IF=enp0s8
ip link set "$IF" up
ip addr replace 10.110.10.10/24 dev "$IF"
mkdir -p /root/A11
cp -a /etc/ssh /root/A11/ssh-before
# TEST-VM konsolida:
# sudo ip link set enp0s8 up
# sudo ip addr replace 10.110.10.30/24 dev enp0s8
```

RDP-SRVda Win+R → ncpa.cpl → lab Ethernet adapter Properties → Internet Protocol Version 4. IP 10.110.10.20, maska 255.255.255.0; default gateway va DNS bo‘sh. MACni hypervisor bilan solishtiring. Windows hostning jismoniy Ethernet/Wi-Fi adapterini o‘zgartirmang.

```
# RDP-SRV: administrator Windows PowerShell 5.1
New-Item -ItemType Directory C:\A11 -Force
Get-NetAdapter | Format-Table Name,MacAddress,Status
Get-NetIPAddress -AddressFamily IPv4
Get-ComputerInfo | Select-Object WindowsProductName,WindowsVersion
hostname
```

1.4. SSH foydalanuvchisi va host kaliti

SSH-SRVda oddiy labssh hisobini yarating; sudo guruhiga qo‘shmang. Mahalliy parolni interaktiv kiriting, bo‘sh parol bermang. Keyingi bosqichda SSH orqali parol autentifikatsiyasi o‘chiriladi, konsol orqali tiklash esa administratorga qoladi.

```
# SSH-SRV root konsoli:
adduser labssh
id labssh
systemctl disable --now ssh.socket
systemctl enable ssh.service
systemctl restart ssh.service
ss -lntp 'sport = :22'
ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub -E sha256
```

Ubuntu 24.04 obrazida SSH socket activation bo‘lishi mumkin. Ushbu variant xizmatni ssh.service orqali boshqaradi, shunda keyingi qayta yuklash tartibi bir xil bo‘ladi. “Unit ssh.socket does not exist” bo‘lsa socket usuli yo‘q; ssh.service holatini tekshirib davom eting.

SHA256 host key fingerprintni VM konsolidan hisobotga qayd eting. Birinchi ulanishda hostda ko‘rsatilgan fingerprint aynan shu qiymatga mos kelishi shart. Ochiq host kalitini qayd etish mumkin; /etc/ssh ichidagi shaxsiy host kalitlarini hisobotga ko‘chirmang.

2. SSH kaliti bilan kirishni sozlash

2.1. Windows hostda Ed25519 kaliti

```
# Windows HOST: oddiy PowerShell; shu sessiyani saqlang.
$a11Dir = Join-Path $env:USERPROFILE 'A11'
New-Item -ItemType Directory $a11Dir -Force | Out-Null
New-Item -ItemType Directory "$env:USERPROFILE\.ssh" -Force | Out-Null
$key = "$env:USERPROFILE\.ssh\a11_ed25519"
$known = "$env:USERPROFILE\.ssh\a11_known_hosts"
if (Test-Path $key) { throw 'A11 kaliti mavjud; qayta yozmang.' }
ssh-keygen -t ed25519 -f $key -C 'A11-labssh'
Copy-Item "$key.pub" "$a11Dir\a11_ed25519.pub"
ssh-keygen -lf "$key.pub" -E sha256
```

Kalit uchun passphrase kiriting. .pub ochiq kalit bo‘lib, serverga ko‘chiriladi; kengaytmasiz a11_ed25519 shaxsiy kalit bo‘lib, hostda qoladi. Passphrase kalitni diskda himoyalaydi; u serverdagi labssh hisobining paroli emas. Kalit avval yaratilgan bo‘lsa yuqoridagi blokni takrorlamang, mavjud A11 kalitini ishlating.

2.2. Ochiq kalitni serverga joylash

```
# SSH-SRV root konsoli:
mkdir -p /mnt/a11share
# O‘z hypervisoringiz uchun bittasini bajaring:
# VMware:
# vmhgfs-fuse .host:/A11 /mnt/a11share -o allow_other
# VirtualBox:
# mount -t vboxsf A11 /mnt/a11share
mountpoint /mnt/a11share
ls -l /mnt/a11share/a11_ed25519.pub
install -d -m 700 -o labssh -g labssh /home/labssh/.ssh
install -m 600 -o labssh -g labssh \
 /mnt/a11share/a11_ed25519.pub /home/labssh/.ssh/authorized_keys
ssh-keygen -lf /home/labssh/.ssh/authorized_keys -E sha256
```

Serverdagi foydalanuvchi kaliti fingerprinti hostdagi .pub fingerprintiga mos bo‘lsin. Bu qiymat 1.4-bo‘limdagi server host key fingerprintidan farq qiladi. Shared folder ishlamasa oldindan tayyorlangan nusxalash yo‘lini tuzating; shaxsiy kalitni serverga ko‘chirmang.

2.3. Cheklov kiritishdan oldin kalitni sinash

```
# Windows HOST:
$sshArgs = @('-i',$key,'-o','IdentitiesOnly=yes',
 '-o',"UserKnownHostsFile=$known",'-o','HostKeyAlgorithms=ssh-ed25519')
ssh @sshArgs labssh@10.110.10.10 'whoami; hostname; id'
```

Birinchi ulanishdagi server fingerprintini VM konsolidagi bilan solishtiring; mos bo‘lsa tasdiqlang. Natijada labssh va server nomi chiqishi kerak. “Enter passphrase for key” mahalliy kalitga tegishli; “labssh@... password” chiqsa kalit bilan kirish ishlamagan, keyingi bloklash bosqichiga o‘tmang. [2]

2.4. SSH cheklovlari — SSH-SRV

Kalit ishlagach, VM konsolida quyidagi konfiguratsiyani yozing. OpenSSHda ko‘p parametrlar uchun birinchi topilgan qiymat amal qiladi; fayl nomining tartibi va Match bloklari ta’sirini sshd -T bilan tekshirish zarur. [2, 3]

```
cat > /etc/ssh/sshd_config.d/00-a11-lab.conf <<'CONF'
PubkeyAuthentication yes
AuthenticationMethods publickey
PasswordAuthentication no
KbdInteractiveAuthentication no
PermitRootLogin no
AllowUsers labssh
MaxAuthTries 3
LoginGraceTime 30
DisableForwarding yes
X11Forwarding no
PermitTunnel no
LogLevel VERBOSE
CONF
/usr/sbin/sshd -t
/usr/sbin/sshd -T -C user=labssh,host=admin,addr=10.110.10.1 \
  > /root/A11/sshd-effective.txt
cat /root/A11/sshd-effective.txt
```

Tekshiriladigan qiymatlar: authenticationmethods publickey, passwordauthentication no, kbdinteractiveauthentication no, permitrootlogin no, allowusers labssh va disableforwarding yes. Natija mos kelmasa /etc/ssh/sshd_configdagi Include tartibi, boshqa .conf fayllari va Match bloklarini tekshiring; faqat o‘zingizning lab obrazingizdagi ziddiyatni tuzating.

2.5. Manba IP bo‘yicha UFW cheklovi

Toza SSH-SRVda avvaldan umumiy 22/TCP yoki OpenSSH allow qoidasi bo‘lmasin. ufw status numbered bilan tekshiring; mavjud keng ruxsatni faqat shu labga tegishli ekanini bilgan holda olib tashlang. Quyidagi tor ruxsat keng ruxsatni avtomatik bekor qilmaydi.

```
ufw status numbered
ufw default deny incoming
ufw default allow outgoing
ufw allow in on "$IF" from 10.110.10.1 to 10.110.10.10 \
  port 22 proto tcp comment 'A11 admin SSH'
ufw --force enable
/usr/sbin/sshd -t && systemctl reload ssh.service
ufw status verbose
systemctl is-active ssh.service
```

VM konsolini ochiq qoldiring. Hostda 2.3-buyruqni yangi ulanish sifatida takrorlang: labssh kalit bilan kirishi kerak. Xizmat portini almashtirish ushbu variantga kiritilmagan; asosiy himoya kalit, ruxsat, manba cheklovi va tekshiruvdan iborat.

2.6. Parol usulini rad etish sinovi — Windows host

```
ssh -o "UserKnownHostsFile=$known" -o PubkeyAuthentication=no `
 -o PreferredAuthentications=password,keyboard-interactive `
 labssh@10.110.10.10
```

Kutiladigan natija: Permission denied (publickey) yoki mos rad javobi. Server parol usulini taklif qilmasligi kerak; bu sinovda parol taxmin qilish talab qilinmaydi. Muvaffaqiyatsiz ulanishning o‘zi yetarli emas: sshd-effective.txtdagi autentifikatsiya qiymatlari ham dalil bo‘lsin.

3. RDP foydalanuvchilari va kirish chegarasi

3.1. Alohida hisoblar — RDP-SRV konsoli

Administrator Windows PowerShell 5.1 oynasini oching. Ikkita yangi mahalliy hisobga turli, noyob va bo‘sh bo‘lmagan parol bering. labrdp ruxsatli oddiy foydalanuvchi, labdeny esa to‘g‘ri parolga ega, ammo RDP huquqi berilmagan nazorat hisobi. Hisoblarni Administrators guruhiga qo‘shmang.

```
$p1 = Read-Host 'labrdp uchun noyob parol' -AsSecureString
New-LocalUser -Name labrdp -Password $p1 -Description 'A11 RDP user'
$p2 = Read-Host 'labdeny uchun boshqa parol' -AsSecureString
New-LocalUser -Name labdeny -Password $p2 -Description 'A11 deny test'
$users = (Get-LocalGroup -SID 'S-1-5-32-545').Name
$rdu = (Get-LocalGroup -SID 'S-1-5-32-555').Name
Add-LocalGroupMember -Group $users -Member labrdp,labdeny
Add-LocalGroupMember -Group $rdu -Member labrdp
Get-LocalGroupMember -Group $rdu
Get-LocalGroupMember -SID 'S-1-5-32-544'
```

Guruhlar SID orqali topilgani uchun Windows interfeysi ruscha yoki inglizcha bo‘lishi mumkin. RDU guruhida labrdp bor, labdeny yo‘q; Administratorsda ikkalasi ham yo‘q bo‘lsin. New-LocalUser “mavjud” desa boshqa hisobni o‘zgartirmang — A11-clean holatidan boshlang.

3.2. Avval tor firewall qoidasini tayyorlash

RDP-SRVdagi lab adapter nomini Get-NetAdapter bilan aniqlang; quyidagi Ethernet qiymatini moslang. Standart RemoteDesktop qoidalari o‘chiriladi, yangi qoida faqat host 10.110.10.1 dan 3389/TCPga ruxsat beradi. UDP ruxsati kiritilmaydi; bu mashqda RDP TCP orqali ishlaydi. [4]

```
$labNic = 'Ethernet'
Get-NetFirewallRule -Name 'RemoteDesktop*' -ErrorAction SilentlyContinue |
 Disable-NetFirewallRule
Set-NetFirewallProfile -Profile Domain,Private,Public -Enabled True `
 -DefaultInboundAction Block
New-NetFirewallRule -Name A11-RDP-Admin -DisplayName 'A11 RDP admin' `
 -Direction Inbound -Action Allow -Protocol TCP -LocalPort 3389 `
 -LocalAddress 10.110.10.20 -RemoteAddress 10.110.10.1 `
 -InterfaceAlias $labNic -Profile Any
Get-NetFirewallRule -Name A11-RDP-Admin | Get-NetFirewallAddressFilter
```

Toza VMda boshqa umumiy 3389 ruxsati bo‘lmasligi kerak. wf.msc → Inbound Rules orqali keng custom qoidalarni ham tekshiring: tor qoida mavjud keng allow qoidasini cheklamaydi. Bu buyruqlar jismoniy Windows hostda bajarilmaydi.

3.3. RDP, NLA va sessiya siyosatini sozlash

Quyidagi blok RDP-SRV konsolida bajariladi. NLA talab qilinadi, TLS security layer tanlanadi, clipboard va disklarni masofaviy sessiyaga ulash taqiqlanadi. So‘ng VM qayta yuklanadi; konsolga administrator bilan qayting. [1, 5]

```
$ts = 'HKLM:\SYSTEM\CurrentControlSet\Control\Terminal Server'
$pol = 'HKLM:\SOFTWARE\Policies\Microsoft\Windows NT\Terminal Services'
Set-ItemProperty -Path $ts -Name fDenyTSConnections -Value 0
New-Item -Path $pol -Force | Out-Null
New-ItemProperty -Path $pol -Name UserAuthentication -PropertyType DWord `
 -Value 1 -Force | Out-Null
New-ItemProperty -Path $pol -Name SecurityLayer -PropertyType DWord `
 -Value 2 -Force | Out-Null
New-ItemProperty -Path $pol -Name fDisableClip -PropertyType DWord `
 -Value 1 -Force | Out-Null
New-ItemProperty -Path $pol -Name fDisableCdm -PropertyType DWord `
 -Value 1 -Force | Out-Null
auditpol /set /subcategory:"{0cce9215-69ae-11d9-bed3-505054503030}" /success:enable /failure:enable
gpupdate /target:computer /force
Restart-Computer
```

GUI orqali tekshirish: gpedit.msc → Computer Configuration → Administrative Templates → Windows Components → Remote Desktop Services → Remote Desktop Session Host. Security bo‘limida NLA talab qilish va specific security layer (SSL); Device and Resource Redirection bo‘limida clipboard va drive redirectionni taqiqlash sozlamalari mavjud. Registry usuli bilan kiritilgan qiymat gpeditda har doim Enabled deb aks etmasligi mumkin; amaldagi qiymat va yangi sessiya natijasi tekshiriladi.

3.4. Qayta yuklangach amaldagi holat

```
# RDP-SRV: administrator PowerShell
$pol = 'HKLM:\SOFTWARE\Policies\Microsoft\Windows NT\Terminal Services'
Get-ItemProperty $pol | Select-Object UserAuthentication,SecurityLayer,
 fDisableClip,fDisableCdm
Get-CimInstance -Namespace root/cimv2/terminalservices `
 -ClassName Win32_TSGeneralSetting -Filter "TerminalName='RDP-tcp'" |
 Select-Object TerminalName,UserAuthenticationRequired,SecurityLayer,
 SSLCertificateSHA1Hash
Get-NetTCPConnection -State Listen -LocalPort 3389
Get-NetFirewallRule -Name A11-RDP-Admin | Get-NetFirewallAddressFilter
```

Siyosatda 1, 2, 1, 1 qiymatlari kutiladi; listener uchun NLA talab qilingan bo‘lsin. SecurityLayer=2 eski TLS 1.0ni majburlash degani emas; protokol versiyalari Schannel va OTga bog‘liq. TLS/NLA muammosini hal qilish uchun eski protokollarni yoki NLAni o‘chirib qo‘ymang; yangilangan obrazni ishlating.

4. Server shaxsini tekshirish va RDP sinovlari

4.1. Laboratoriya RDP sertifikati

Domen va CA yo‘q toza Windows VM odatda o‘z-o‘zini imzolagan RDP sertifikatidan foydalanadi. Bunday sertifikat shifrlashni ta’minlashi mumkin, lekin mijozda oldindan ishonch zanjiri mavjud emas. Avval RDP-SRVning VM konsolidan sertifikat tafsilotlarini oling. [6]

```
# RDP-SRV konsoli:
Get-ChildItem 'Cert:\LocalMachine\Remote Desktop' |
 Select-Object Subject,Thumbprint,NotBefore,NotAfter |
 Format-List | Out-File C:\A11\rdp-certificate.txt
Get-Content C:\A11\rdp-certificate.txt
```

Windows hostda mstsc ni oching, manzil sifatida 10.110.10.20 ni kiriting. Sertifikat ogohlantirishi chiqsa View certificate → Details orqali Subject, amal muddati va Thumbprintni konsoldagi bilan solishtiring. Bir nechta sertifikat bo‘lsa, amalda taklif qilingan sertifikatni aniqlang. Mos kelmasa ulanishni to‘xtatib, IP va VMni tekshiring.

Faqat shu ajratilgan laboratoriyada, barcha tafsilotlar konsoldan tasdiqlangach, ushbu ulanish uchun davom etish mumkin. “Don’t ask me again”ni belgilamang va server sertifikatini global ishonch omboriga qo‘shmang. Haqiqiy tashkilotda nomi mos, ishonchli CA bergan sertifikat va uning yangilanish tartibi kerak. Thumbprintni taqqoslash CA boshqaruvini almashtirmaydi.

4.2. Ruxsatli foydalanuvchi bilan kirish

- Windows host: Test-NetConnection 10.110.10.20 -Port 3389. TcpTestSucceeded=True faqat portga yetib borishni ko‘rsatadi.
- mstsc → Show Options → User name: RDP-SRVning haqiqiy kompyuter nomi\labrdp. Kompyuter nomini serverdagi hostname natijasidan oling; hostning foydalanuvchi nomini kiritmang.
- Local Resources → Clipboardni belgilangan holda qoldiring; More… → bitta host diskini sinov uchun belgilang. Parolni saqlashni tanlamang.
- RDP sessiya ochilgach oddiy terminalda whoami bajaring; labrdp ekanini qayd eting. Administrator huquqini talab qiluvchi sozlamalarni o‘zgartirmang.
- Hostdagi Notepaddan A11-clipboard-test matnini nusxalab, RDP Notepadga qo‘yishga urining. Matn o‘tmasligi; This PCda hostdan yo‘naltirilgan disk ko‘rinmasligi kutiladi. Faqat yangi sessiyada sinang.

Clipboard sinovida boshqa clipboard menejeri yoki hypervisorning konsol almashish funksiyasidan foydalanmang; aynan RDP oynasidagi yo‘l tekshiriladi. Server siyosati va mijozda yo‘naltirish so‘ralgani birgalikda dalil bo‘lsin.

4.3. Huquqi yo‘q foydalanuvchi

Avval labrdp sessiyasidan Sign out qiling. Hostda yangi mstsc ochib kompyuter_nomi\labdeny hisobini va uning to‘g‘ri parolini bir marta kiriting. Masofaviy kirish huquqi yo‘qligi sabab sessiya ochilmasligi kerak. Noto‘g‘ri parol bilan sinash bu huquq cheklovini isbotlamaydi. Rad matni Windows versiyasiga qarab farq qiladi.

5. Manba cheklovi, loglar va natija jadvali

5.1. Ruxsatsiz manba — TEST-VM

```
# TEST-VM: faqat ikki aniq IP/portga ulanish sinovi.
ip -br addr
ip route get 10.110.10.10
nc -vz -w 3 10.110.10.10 22
nc -vz -w 3 10.110.10.20 3389
```

Kutiladigan natija: ikkala TCP ulanishi ham o‘tmaydi, odatda timeout. TEST-VMning manbasi 10.110.10.30 bo‘lsin. So‘ng hostda ikkala xizmatga qayta muvaffaqiyatli kiring: bu salbiy natija xizmat o‘chganidan emas, manba cheklovidan kelganini ko‘rsatadi.

5.2. SSH logi — SSH-SRV konsoli

```
journalctl -u ssh.service --since '90 minutes ago' --no-pager \
  > /root/A11/ssh-journal.txt
cat /root/A11/ssh-journal.txt
ufw status numbered > /root/A11/ufw-status.txt
```

Accepted publickey yozuvida labssh, manba IP va kalit fingerprintini toping. Parol usuli o‘chirilgan sinovdagi yozuvlar versiyaga qarab farq qilishi mumkin. Firewall tashlagan urinish sshdga yetmasligi sabab SSH logida ko‘rinmasligi mumkin; bo‘sh ilova logi urinish bo‘lmaganini isbotlamaydi.

5.3. Windows kirish hodisalari — RDP-SRV

```
Get-WinEvent -FilterHashtable @{LogName='Security';Id=4624,4625;
 StartTime=(Get-Date).AddMinutes(-90)} -ErrorAction SilentlyContinue |
 Select-Object TimeCreated,Id,Message |
 Format-List | Out-File C:\A11\logon-events.txt
Get-Content C:\A11\logon-events.txt
```

Windows hodisalari mahalliy vaqt bilan chiqadi; hisobotda SSH va Windows vaqtlarini bir vaqt zonasiga keltiring. Event Viewer → Windows Logs → Securityda labrdp/labdeny, manba IP va vaqtni solishtiring. 4624 ichidagi Logon Type 10 masofaviy interaktiv kirishga tegishli. NLA jarayonida boshqa turdagi hodisalar ham bo‘lishi mumkin; 4624 sonini RDP sessiyalar soni deb olmang. Rad etilgan kirishda 4625 yoki RDP operatsion loglarini tahlil qiling. [7]

№ / sinov	Kutiladigan natija	Haqiqiy natija / dalil
1. Host → SSH, to‘g‘ri kalit	labssh kiradi	_____
2. Host → SSH, faqat parol usuli	Rad etiladi; publickey talab qilinadi	_____
3. TEST-VM → SSH:22	TCP ulanishi o‘tmaydi	_____
4. Host → RDP, labrdp	Oddiy foydalanuvchi sessiyasi	_____
5. Host → RDP, labdeny + to‘g‘ri parol	Kirish huquqi sabab rad	_____
6. TEST-VM → RDP:3389	TCP ulanishi o‘tmaydi	_____
7. RDP clipboard	Host matni o‘tmaydi	_____
8. RDP disk yo‘naltirish	Host diski ko‘rinmaydi	_____

6. Mustaqil topshiriqlar va nosozliklarni aniqlash

- SSH va RDP uchun alohida “talab → sozlama → tekshiruv → dalil” jadvalini tuzing. Autentifikatsiya, ruxsat, server shaxsi va tarmoq chegarasi alohida ko‘rsatilsin.
- 5-bo‘limdagi sakkiz sinovni bajaring; kutilgan va haqiqiy natijani yozing. Har salbiy sinovdan keyin ruxsatli kirish ishlayotganini qayta tasdiqlang.
- SSH host key va foydalanuvchi kaliti fingerprintini taqqoslang; ular nimani isbotlashini izohlang. RDP sertifikatining ishonch cheklovini yozing.
- Kamida 5 izohli skrinshot oling: SSH amaldagi konfiguratsiyasi; muvaffaqiyatli kalitli kirish; RDP NLA va firewall manbasi; labdeny rad javobi; log va TEST-VM natijalari.
- Haqiqiy universitet tarmog‘i uchun uchta takomillashtirish taklif qiling: boshqaruv segmenti/VPN yoki RD Gateway, MFA integratsiyasi, ishonchli sertifikat yoki markaziy hisob siyosati. Har biri qaysi xavfni kamaytirishini yozing.

Muammo	Tekshirish va yechim
SSH: Permission denied	labssh hisob holati, .ssh egasi/700, authorized_keys egasi/600; .pub kalit fingerprinti; sshd -T va journal.
SSH konfiguratsiyasi ta’sir qilmadi	Include va .conf tartibi, Match bloki; sshd -t xatosi; reload muvaffaqiyati; socket/service holati.
Host key changed	VM tiklanganmi yoki server almashganmi? Konsoldan yangi fingerprintni tasdiqlang; tekshirmasdan known_hostsni tozalamang.
Host ham portga yetmaydi	Host-only adapter IPsi, manba marshruti, service listener, firewall scope va lab adapter nomi.
TEST-VM portga kira oladi	Oldingi keng allow qoidasi, noto‘g‘ri adapter, host-onlydan tashqari yo‘l, manba IPsi; firewall yoqilganmi?
RDP-SRVda RDP yo‘q	Windows Home emasligini tekshiring; Pro/Education/Enterprise obrazini qo‘llang.
labrdp ham kira olmaydi	Parol/PIN farqi, kompyuter_nomi\labrdp, RDU guruhi va secpol.msc kirish huquqlari; Deny huquqi ustun.
Clipboard yoki disk o‘tmoqda	Yangi RDP sessiyasi, fDisableClip/fDisableCdm=1, tashqi GPO ta’siri va aynan RDP kanali ishlatilganini tekshiring.

Konsoldan tiklash

SSH sozlamasi noto‘g‘ri bo‘lsa, VM konsolida faqat 00-a11-lab.conf faylini vaqtincha /root/A11ga ko‘chiring, sshd -tni qayta bajaring va ssh.service ni reload qiling. Firewall scope xatosini konsoldan to‘g‘rilang. RDPda administrator konsoli orqali siyosat va firewallni tuzating. Eng toza qaytish usuli — dalillar saqlangach A11-clean snapshot.

7. Hisobot, vaqt taqsimoti va baholash

7.1. Hisobot tuzilishi

Titul: fan, A11, 1-variant, F.I.Sh., guruh va sana. Asosiy qism: maqsad; topologiya va IP reja; OT/dastur versiyalari; SSH/RDP siyosat jadvallari; sakkiz sinov natijasi; izohli skrinshotlar; log tahlili; 5–7 jumlalik xulosa. Muvaffaqiyatli ulanishning o‘zi barcha himoya ishlashini isbotlamasligini tushuntiring.

Ilovalar: SSH amaldagi konfiguratsiyasi, UFW holati va tegishli journal parchasi; RDP siyosat, firewall manbasi, sertifikatning ochiq tafsilotlari va lab hisoblariga tegishli hodisalar. Parol, passphrase, shaxsiy SSH kaliti yoki butun /etc/ssh zaxirasini ilova qilmang.

```
# SSH-SRV: faqat hisobot uchun kerakli ochiq dalillar.
cp /etc/ssh/sshd_config.d/00-a11-lab.conf /root/A11/
cd /root/A11
tar -czf A11_ssh_dalillar.tar.gz 00-a11-lab.conf \
  sshd-effective.txt ufw-status.txt ssh-journal.txt
mountpoint /mnt/a11share && cp A11_ssh_dalillar.tar.gz /mnt/a11share/
```

RDP-SRVdagi C:\A11 dalillarini oldindan sozlangan shared folder orqali hostga ko‘chiring. RDP disk yo‘naltirishni fayl chiqarish uchun qayta yoqmang. Windows Security logidan hisobotga faqat tegishli o‘quv hisoblari va vaqt oralig‘ini ajrating.

7.2. Tavsiya etilgan 80 daqiqalik reja

Bosqich	Daqiqa
Muhit, manzillar va xizmatlarni tekshirish	10
SSH kaliti va dastlabki kirish	15
SSH cheklovlari, UFW va sinovlar	10
RDP hisoblari, firewall, NLA va qayta yuklash	15
Sertifikat, RDP huquqi va yo‘naltirish sinovlari	15
TEST-VM, loglar va hisobot/himoya	15
Jami	80

7.3. Baholash mezonlari — 10 ball

Mezon	Ball
Muhit, IP va alohida oddiy foydalanuvchilar	1
SSH kaliti, host fingerprint va samarali konfiguratsiya	2
SSH salbiy sinovi va manba cheklovi	1
RDP NLA/TLS, foydalanuvchi va firewall scope	2
RDP huquqi, manba, clipboard/disk sinovlari	2
Izohli loglar, hisobot va og‘zaki himoya	2
Jami	10

Mezon ichida to‘liq ishlash dalili — to‘liq ball, qisman natija — tegishli ulush. Kutilgan holatni haqiqiy natija sifatida ko‘rsatish mumkin emas. Nosozlik sababini dalil bilan aniqlash tahlil bahosida hisobga olinadi.

8. Nazorat savollari va yakunlash

1. SSH host key bilan foydalanuvchi public key o‘rtasidagi farq nima?
2. Passphrase va server foydalanuvchisi paroli qayerda tekshiriladi?
3. Nega SSH kaliti bilan kirishni tekshirmasdan parol usuli o‘chirilmaydi?
4. sshd -t va sshd -T nimani tekshiradi?
5. Nima uchun tor firewall allow qoidasi keng allow qoidasini bekor qilmaydi?
6. RDPda NLA, TLS va MFA vazifalari qanday farqlanadi?
7. labdeny sinovida nima uchun to‘g‘ri parol ishlatiladi?
8. O‘z-o‘zini imzolagan sertifikat bilan ishonchli CA sertifikati qanday farqlanadi?
9. 4624 hodisasining o‘zi RDP sessiyasi ochilganini qachon yetarli ko‘rsatmaydi?
10. RDP clipboard cheklovi barcha fayl chiqarish kanallarini to‘xtatadimi?

8.1. Ishni yakunlash

Hisobot va dalillar hostda ochilishini tekshiring. RDPdan Sign out, SSHdan exit qiling, VMlarni o‘chiring va A11-clean snapshotga qayting. Hostdagi keyingi ishlarga aralashmasligi uchun faqat A11ga tegishli shared folder ulanishini uzing. Keraksiz A11 kaliti va a11_known_hostsni o‘qituvchi belgilagan saqlash muddati tugagach olib tashlang; boshqa kalit/known_hostsga tegmang.

8.2. Foydalanilgan manbalar

[1] Microsoft Learn — Enable Remote Desktop; Windows nashrlari va NLA.
https://learn.microsoft.com/en-us/windows-server/remote/remote-desktop-services/remotepc/remote-desktop-allow-access

[2] Ubuntu Server — OpenSSH server: o‘rnatish, kalit va konfiguratsiya.
https://ubuntu.com/server/docs/how-to/security/openssh-server/

[3] OpenBSD/OpenSSH — sshd_config(5).
https://man.openbsd.org/sshd_config

[4] Microsoft Learn — Windows Firewall Rules.
https://learn.microsoft.com/en-us/windows/security/operating-system-security/network-security/windows-firewall/rules

[5] Microsoft Learn — ADMX_TerminalServer siyosat parametrlari.
https://learn.microsoft.com/en-us/windows/client-management/mdm/policy-csp-admx-terminalserver

[6] Microsoft Learn — Remote Desktop listener certificate configurations.
https://learn.microsoft.com/en-us/troubleshoot/windows-server/remote/remote-desktop-listener-certificate-configurations

[7] Microsoft Learn — Event 4624: muvaffaqiyatli kirish va logon turlari.
https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-10/security/threat-protection/auditing/event-4624

Mavzu foydalanuvchi taqdim etgan “Tarmoq xavfsizligi fani mavzulari.docx” hujjatidagi A11 bandiga mos. Tuzilish “Kompyuter_tarmoqlari_A6_1_variant_batafsil.docx” namunasiga moslashtirilgan. O‘qituvchi darsdan oldin bir hostda to‘liq senariyni tekshiradi va tayyor obrazlarni tarqatadi.